

BOOK REVIEWS

RAMANUJAN: LETTERS AND COMMENTARY (History of Mathematics 9)

By BRUCE C. BERNDT and ROBERT A. RANKIN: 347 pp., US\$79.00,
LMS Members' price US\$47.00, ISBN 0 8218 0287 9 (American Mathematical
Society and London Mathematical Society, 1995).

Srinivasa Ramanujan is one of the most romantic figures in mathematics. The story of the Indian clerk, a self-taught mathematician, producing extraordinary mathematical results which took him to Cambridge only to suffer an early death at the age of 32, is well known. Even non-mathematicians are aware of the tale thanks to an excellent popular biography [3].

The authors have collected as many letters to, from and about Ramanujan as possible, and added a commentary. This includes sketch biographies of the correspondents and, for the more mathematical correspondence, a history of work on the topics mentioned. The result is a book which may be enjoyed at different levels. For the non-specialist it may be read as a supplement to the biography. Mathematicians specializing in areas that arose from Ramanujan's work will be interested to see some of the correspondence with Hardy, developing the original ideas on such topics as modular functions, the Hardy Ramanujan method, and mock theta functions. At this level the book may be regarded as a supplement to Berndt's long-term project of publishing an edited version of Ramanujan's notebooks [1].

Chapter 1 covers the time when the administrators in Madras were beginning to recognize that Ramanujan had extraordinary talents. Unfortunately, the first English mathematician approached, M. J. M. Hill of University College, London, was an applied mathematician who did not fully appreciate what Ramanujan had achieved. Chapter 2 covers the correspondence with G. H. Hardy. Although it is known that Ramanujan had previously written to H. F. Baker and E. W. Hobson, this correspondence has been lost. The letters to Hardy had previously been available only in edited form as an appendix to Ramanujan's collected papers [4]. Here Hardy's replies are also included, but not his comment on Ramanujan's results [2]: 'I had never seen anything in the least like them before. A single look at them is enough to show that they could only be written down by a mathematician of the highest class. They must be true because, if they were not true, no one would have had the imagination to invent them.'

Chapter 3 covers the preparations for coming to England, with various memos devoted to obtaining a University of Madras scholarship, and Hardy's letters trying to impress on Ramanujan the need for rigour. Chapter 4 covers the time at Cambridge, whilst by mid-1917 Ramanujan had entered a nursing hostel. Chapter 5, 'Ramanujan is ill', conveys Hardy's sense of urgency in securing for Ramanujan Fellowships of the Royal Society and Trinity College. Because of his separation from Cambridge, there are several long mathematical letters to Hardy, including work on congruences for the partition function and remarks on sums of squares and Waring's problem.

When Ramanujan returned to India in 1919, the *Madras Times* (5 April 1919) carried a long article describing Ramanujan's career, which is included in full in Chapter 6 (pp. 205–213), making it clear the regard in which Ramanujan's achievements were held in Madras. Chapter 7, 'After his death', contains an article from the *Madras Mail* (27 April 1920) printed the day after Ramanujan's death, and various letters concerned with the pension provisions for his family.

Chapter 8 is concerned with the arrangements for publishing Ramanujan's collected papers and the story of Ramanujan's notebooks. G. N. Watson and B. M. Wilson intended to work through the notebooks, providing an edited version of the second notebook (which is a revised, enlarged version of the first), with proofs of Ramanujan's statements. Although they made considerable progress on this, the project was never completed due to Wilson's death in 1935. Subsequently, the notebooks disappeared from view, only to resurface amongst Watson's books and papers after his death in 1965. Chapter 9 contains a brief family history, and finally there is a comprehensive list of 302 references.

The book is very readable, contains much material not available elsewhere, and can be read at a variety of levels, so it can be highly recommended to anyone with an interest in Ramanujan.

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R. J. COOK

RINGS WITH GENERALIZED IDENTITIES (Pure and Applied Mathematics 196)

By K. I. BEIDAR, W. S. MARTINDALE III and A. V. MIKHALEV: 522 pp.,
US\$185.00, ISBN 0 8247 9325 0 (Dekker, 1996).

Polynomial identities first arose in the algebraic treatment of projective geometry. Any projective plane can be coordinatized by an algebra called a ternary field, which reduces to a skew field if and only if Desargues' theorem holds, while Pappus' theorem is needed to ensure commutativity. These questions were studied in the early part of this century, but it was not until the 1940s that pure PI-theory began to develop in the wake of the study of rings without finiteness assumptions, beginning with Kurosh's question: 'Is every algebraic algebra locally finite?'

Let $F = k\langle X \rangle$ be the free associative k -algebra (where k is a commutative field) on a set X . Any non-zero element of F is called a *polynomial identity* (PI) of a k -algebra A if it is mapped to zero by all homomorphisms from F to A , and an algebra with such a PI is called a *PI-algebra*. A first landmark was Kaplansky's 1948 paper, in which he proved that every primitive algebra satisfying a polynomial identity of degree d is a simple algebra of dimension at most $(d/2)^2$ over its centre. This was followed by a brisk development which received a connected presentation in 1973 by Procesi and a more detailed account by Rowen in 1980.

The polynomial identities defined above were rather special in that only scalar

coefficients (commuting with all the variables) were admitted. To allow more general cases, one replaces the free algebra F by the tensor A -ring $A_k \langle X \rangle$, defined as the ring generated by A and X subject to the relations $cx = xc$ for all $x \in X$, $c \in k$. Now *generalized polynomial identities* (GPIs) of A are defined as non-zero elements of $A_k \langle X \rangle$ mapped to zero under all homomorphisms to A , and algebras possessing them are called *GPI-algebras*. The net is now cast much wider, since every algebra that is not prime satisfies an identity $axb = 0$. Nevertheless, in 1965 Amitsur succeeded in proving a decisive GPI-theorem: a primitive ring R satisfies a GPI if and only if it has a non-zero socle and the endomorphism ring of a simple R -module is finite-dimensional over its centre. The proof was a fairly complicated and highly ingenious induction; a little later, Martindale, one of the authors of the book under review, gave another, simpler proof and at the same time extended the result to prime rings, using the quotient ring construction to obtain a reduction to the primitive case. Since then there has again been much activity, extending results and putting them in a wider context, much of it due to Beidar and Mikhalev, who proved the GPI-theorem for semiprime rings, using the method of orthogonal completion to reduce the problem to the prime case, and also considered rings with a given antiautomorphism.

The present book is a full account of all these methods. After an initial chapter on free algebras and coproducts, the quotient ring construction is described for semiprime rings; the authors follow the explicit method of Utumi, and also show that derivations and (anti)automorphisms can be extended to the quotient ring. There follows a chapter on the orthogonal completion, basically a method for reducing a semiprime orthogonally complete ring to its prime factors by relating each Horn formula to its projections on the factors. After a recap of facts on primitive rings used later in the book, there comes a first application: the Poincaré–Birkhoff–Witt theorem. The reason for devoting a whole chapter to this relatively simple result is that a more general form is proved, referring to differential Lie algebras, which represents a background for some of the later applications.

After these preparations, the authors are able to give a very short proof of the main GPI-theorem for prime rings, essentially Martindale's proof with simplifications due to C.-L. Chuang. From it, the theorems of Kaplansky and Posner are easily obtained (though not quite in their sharpest form). Next, Beidar's extension to semiprime rings is presented, as well as the three authors' theorem that GPIs extend from semiprime rings to their rings of quotients. The GPIs of a given prime or semiprime ring R form a T -ideal (that is, an ideal admitting all endomorphisms) of $R_C \langle X \rangle$; this T -ideal was first considered by D. E. Littlewood, who obtained an explicit identity generating all multilinear identities in $M_n(C)$, the ring of $n \times n$ matrices over a commutative field C . His result was generalized to primitive algebras by Beidar in his PhD Thesis of 1978, and a full account is given here. The main result states that for a primitive algebra R over an infinite field C , with an idempotent e such that eRe is a division algebra with centre eC , the T -ideal of GPIs is generated by a single element, a generalized standard polynomial with e appearing as coefficient between the variables. The techniques introduced earlier lead to extensions to semiprime rings, which, however, are less decisive.

Closely related to GPIs is Kharchenko's work on T -identities. The algebra now comes equipped with a set of derivations and automorphisms and a T -identity is a (non-zero) polynomial involving, besides the variables and coefficients, the given automorphisms and derivations. Now there is no longer an obvious normal form, but a reduction process is described leading to such a form. Identities in this form are

called *reduced*, and Kharchenko's theorem is proved, stating that a prime ring with a reduced T -identity holding on a non-zero ideal is itself a GPI-ring. There is a fuller investigation of T -identities, leading to simple proofs of various well-known results on algebraic automorphisms and derivations. Much of this work can, at the cost of some complication, also be carried over to semiprime rings.

In 1961 Herstein raised several questions about Lie algebras which have given rise to much activity. If R is a simple ring with involution and K is the Lie ring of skew elements, is every automorphism of K induced by an automorphism of R (up to a possible scalar term)? Over the years, partial answers to this and a similar question on Lie automorphisms of simple rings have been given, but a full solution was not found until 1993, when the latter problem was solved by Brešar. Here the authors present an almost complete solution of the problem stated, giving sufficient conditions for a Lie isomorphism between the skew elements of prime rings with involution to extend to an isomorphism of the associative subrings generated. The proof, involving some calculation but little of the theory developed in earlier pages, throws some light on the situation but does not really point the way to a complete answer.

We have here an account starting from the central questions of the subject, of interest to all ring theorists, and going to peripheral problems with rather more limited appeal. What makes the book of wider interest is the account of reduction techniques, perhaps not as well known as they should be, and certainly applicable in many other situations. The book has a full bibliography but only a rudimentary index, though the list of symbols is helpful. Unfortunately, the presentation is rather convoluted, often indulging in needless complications and irritating slips of grammar. It also comes at a price few individuals will wish to pay. However, it is without question a book of importance, which should not be missing from any mathematics library.

P. M. COHN

REPRESENTATION THEORY OF ARTIN ALGEBRAS (Cambridge Studies in Advanced Mathematics 36)

By MAURICE AUSLANDER, IDUN REITEN and SVERRE O. SMALØ: 423 pp., £50.00,
ISBN 0 521 41134 3 (Cambridge University Press, 1995).

This is one of the first systematic, approachable and comprehensive accounts of an area of representation theory which took off in the 1960s and which was the subject of rapid development during the next decades. The authors faced the problem of imposing order in an account of a subject where there was no adequate account before, and of setting it down in a manner which is complete, definitive and accessible. What they have produced is the best single account of the representation theory of Artin algebras currently available.

The part of this theory described in this book has to do with representations of quivers, and with almost split sequences—the latter invented by Auslander and Reiten. These new tools in representation theory have the potential to change the way finite-dimensional algebras are taught in universities, since they are fundamental, of broad interest, and can be explained in a quite elementary way. For example, a quiver is just a directed graph, and a representation of it consists of the specification of a

vector space (over some field) for each vertex of the quiver, together with a linear map for each directed edge (or arrow) in the quiver, going between the vector spaces associated to the end vertices of the arrow. There is a notion of a homomorphism of representations, to do with a commutative diagram of linear maps, and hence of isomorphism of representations. One may now study the structure of such representations, ultimately trying to classify them up to isomorphism.

This idea seems simple enough, but it may be hard to see why one should ever want to consider such things. Classifying representations of quivers is really a formalization of the problem of classifying sets of matrices up to conjugacy. The case of classifying a single endomorphism of a vector space (over an algebraically closed field) is solved by the Jordan canonical form, in that two endomorphisms are conjugate if and only if they have the same Jordan canonical form. In terms of quivers, this problem is the same as classifying the representations of the quiver with a single vertex and a single directed edge forming a loop from that vertex to itself. More generally, whenever one has a set of operators acting on some space satisfying a fixed set of relations, the problem of classifying such actions is the same as classifying the representations of an algebra generated abstractly by the operators, and this is the same as classifying the representations of a corresponding quiver (with relations). This idea is so basic that it should be of interest to people outside the area of pure representation theory: for example, to physicists.

As for the almost split sequences and the related Auslander–Reiten quiver, these also may be described in an elementary, axiomatic way. They are related to the notion of an irreducible morphism, which can be regarded as a generalization of an irreducible element in a ring. The Auslander–Reiten quiver turns out to have remarkable regularity properties, and its structure has been a main ingredient in a whole range of theorems to do with characterizing finite representation type, and classifying various classes of algebras. These properties and many of the applications are carefully developed in this book.

One hopes for so many things from a book such as this, since it is one of the first expositions of a recent theory. Ideally, it should be broad in its coverage, complete, carefully organized, definitive, self-contained, and suitably elementary but at the same time deep and sophisticated. This daunting set of requirements is more than it is reasonable to ask to be satisfied, and it is of course possible to point to areas where more could have been done. The treatment is carefully organized and definitive. Much of the time it is deep and sophisticated. Out of necessity in a book of reasonable size, it is not complete (a list of topics the book does not treat is given in the introduction). These last two aspects unfortunately mean that the book is not as well suited for use as a text in a lecture course as it might be. It begins at a reasonably advanced level (assuming some basic properties of the radical, projective and injective modules, and what a category is, for example), and because of the sophistication of the ideas it would not be easy for students to start this book without a fairly strong preparation in algebra. In Britain this means that the book would be suitable for use only at the post-graduate level, and in the United States it would not be suitable for use until perhaps the final part of the first year of graduate school. This is a pity because, as I have pointed out, many of the ideas are quite elementary and can be appreciated at an earlier stage than this.

This book will be the major source for students studying in this field, and a reference for the material it covers. It is a great pity that Maurice Auslander did not live to see the publication of this work, in the area in which, with his co-workers (and

very notably Idun Reiten and Sverre Smalø), he remained a leader and a pioneer to the end of his career. It is also a pity that he is not here to work on the ‘forthcoming volume dealing with other aspects of the subject’ referred to in the introduction to the book. This future book is also much needed.

PETER WEBB

COMPLEX DYNAMICS AND RENORMALIZATION

(Annals of Mathematics Studies 135)

By CURTIS T. McMULLEN: 214 pp., £35.00, ISBN 0 691 02981 4
(Princeton University Press, 1995).

The amazingly intricate and beautiful self-similar structure of the Mandelbrot set

$$M = \{c : f_c^n(0) \not\rightarrow \infty \text{ as } n \rightarrow \infty\} \quad (f_c(z) = z^2 + c),$$

apparent in so many remarkable computer pictures, has associated to it equally extraordinary and rich mathematics combining complex analysis, hyperbolic geometry and symbolic dynamics. The unifying theme of the wave of mathematical development uncovering this structure over the last 15 years has been the viewpoint of *renormalization*.

Curt McMullen’s book fulfils two objectives: it is a research monograph, concluding with the proof of a major new result (see below), and at the same time it provides a superb, if rapid, introduction to many of the tools of the new subject and to the conjectures that are the driving force behind their development, conjectures which remain tantalizingly just out of reach in their full generality (a good summary of the present situation can be found in [7]).

To state McMullen’s main result in context, we start by recalling that in 1982, in the early days of the study of the Mandelbrot set M , Douady and Hubbard gave a brief and elegant proof [4] that M is connected, by constructing an explicit conformal bijection Ψ between the complement of M in the Riemann sphere and the (open) unit disc D . They went on in [5] to discuss the conjecture that the boundary ∂M of M is *locally* connected, from which it would follow that Ψ^{-1} would extend to a continuous surjection $\partial D \rightarrow \partial M$, reducing the study of the structure of M to combinatorics. This conjecture (known as ‘MLC’) has stubbornly resisted proof in its full generality, despite being proved by Yoccoz to hold at all bar the ‘infinitely renormalizable’ $c \in \partial M$ and by Lyubich to hold at a large sub-class of these remaining points. One of the consequences of MLC, as is shown in [5], would be that for every value of c in the interior M° of the Mandelbrot set, the map f_c would be *hyperbolic*, in the sense that the orbit of the critical point 0 would tend to an attractive periodic cycle. The latter conjecture (‘ M° is hyperbolic’) is of considerable interest in its own right, being the special case for quadratic polynomials of what is perhaps the main conjecture of conformal dynamics, that hyperbolic maps form an open dense subset in the space of degree d rational maps of the Riemann sphere. McMullen’s book culminates with his proof that all components of M° meeting the real line are hyperbolic, a major new result (also proved independently by Lyubich and by Swiatek) with applications to real dynamics as the intensively studied *logistic family* of maps of the real unit interval, $f_\mu(x) = \mu x(1-x)$, is simply a reparametrization of the quadratic family f_c .

En route to this theorem, McMullen introduces to the reader the wonderful range of tools at the disposal of the modern holomorphic dynamicist. Topics discussed include the Koebe distortion theorems for univalent maps, the collar theorem for geodesics on surfaces, classical work of Fatou and Julia on iterated rational maps, the Douady–Hubbard theory of polynomial-like mappings, ergodic theory of rational maps, invariant line fields, holomorphic motions and stability in families of rational maps, the Mandelbrot set and Thurston’s lamination model, Riemann mappings and external rays, renormalization, Yoccoz puzzles and Branner–Hubbard tableaux, Sullivan’s *a priori* bounds, and finally the application of all this machinery to the proof of rigidity theorems. Appendices discuss orbifolds associated to rational maps, and the notion of a *quotient* of a rational map, a notion that allows the author to state and prove a *closing lemma* for rational maps (introducing Thurston’s characterization of critically finite rational maps on the way). The new ideas come thick and fast, but the whole work is bound together by the author’s drive towards the hyperbolicity conjecture for quadratic maps, via the conjecture (which he proves to be equivalent) that a quadratic map carries no invariant line field on its Julia set.

McMullen’s book is a very welcome and timely addition to the literature. Its main results demonstrate the power of the Sullivan–McMullen approach to rigidity in conformal dynamics (another example of which is provided by Sullivan’s proof of Feigenbaum universality, presented in another impressive recent book [8]). The pace is swift but the author has chosen his route through the subject with care and skill, and the presentation is clear and accurate, with references to more detailed sources where appropriate. I know of no other text which takes the proverbially well-prepared graduate student right up to the forefront of recent developments in holomorphic dynamics. But it moves very quickly, and the aforementioned student would be well-advised to prepare herself or himself by first reading one of the texts [1, 3, 10], which cover the classical theory of Fatou and Julia, and some of the more recent developments; the survey articles [2] and [6], and lecture notes [9], also provide a very good initiation into complex iteration theory.

The story of complex dynamics is, of course, still far from complete. Not only are the main conjectures concerning the Mandelbrot set M still unresolved, but every new result seems to point to new levels of complexity in the behaviour of polynomial maps: two striking recent examples are Shishikura’s proof that ∂M has Hausdorff dimension 2, and that of Nowicki and van Strien of the existence of polynomials of the form $z \rightarrow z^{2^n} + c$ (with n large) having Julia sets of positive area. As the story unfolds, weightier and more comprehensive texts may appear, but McMullen’s monograph will surely come to be regarded as a classic, both for its proof of one of the major results of complex renormalization theory, and as a very readable route into the subject.

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SHAUN BULLETT

MIXED PROBLEM FOR PARTIAL DIFFERENTIAL EQUATIONS WITH QUASIHOMOGENEOUS PRINCIPAL PART

(Translations of Mathematical Monographs 147)

By S. G. GINDIKIN and L. R. VOLEVICH: 232 pp., US\$99.00, ISBN 0 8218 4617 5
(American Mathematical Society, 1996).

There is an enormous number of books devoted to mixed problems for linear partial differential equations. The subject has been studied in great detail, and one does not usually want to look at yet another book where the same results are proved under slightly less restrictive conditions or for a bit more general classes of operators. However, the book under review has advantages which make it distinctive.

First, the authors consider much more general classes of evolutionary equations than one normally does. The type of equation is determined by the main part of the symbol of the corresponding differential operator. Mixed problems are normally studied for parabolic and hyperbolic equations, the most famous examples of which are

$$\partial_t u - \sum_j \partial_{x_j}^2 u = f \quad (\text{the heat equation})$$

and

$$\partial_t^2 u - \sum_j \partial_{x_j}^2 u = f \quad (\text{the wave equation}).$$

The book deals with operators whose symbols are quasihomogeneous. For such operators, even the definition of the main part of the symbol is not quite trivial, and it is not obvious which terms in the differential expression should be regarded as those of lower order. The problem of finding appropriate boundary conditions is even more difficult.

Secondly, the authors use a modern version of the classical energy method which proves to be one of the most powerful tools for the study of linear evolutionary equations and corresponding mixed problems. This method was originally developed for the hyperbolic equations. The authors show that it is almost automatically extended to parabolic equations and, more generally, to operators with quasihomogeneous symbols. In a sense, the authors have constructed a unified theory of mixed problems for linear partial differential operators with dominant principal quasihomogeneous part. They have also obtained some results for systems of differential operators of this type.

Of course, the most important classes of operators with quasihomogeneous symbols (like $2b$ -parabolic operators) have been studied for a long time, and most of

the results in the book are already known. The main contribution of these authors is concerned with the so-called $(2b+1)$ -hyperbolic operators. The simplest example of such operators is

$$\pm \partial_t - \sum_j \partial_{x_j}^3$$

(in dimension one it is the linear part of the KdV operator). The authors develop a theory of mixed problems for such operators which may be of interest to specialists working in relevant fields.

However, even if the book did not contain any new results, it would deserve to be published. It gives a good introduction to the subject, and can be understood by undergraduate students (in particular, it contains a brief summary of the necessary results from the theory of distributions). The fact that the authors apply the energy method in high generality makes its essence even more clear, and an expert may well find many useful technical results and observations that were previously unknown.

YU. SAFAROV

DIRICHLET FORMS AND SYMMETRIC MARKOV PROCESSES (Studies in Mathematics 19)

By MASATOSHI FUKUSHIMA, YOSHIDA OSHIMA and MASAYOSHI TAKEDA: 392 pp.,
DM.168.-/US\$79.95, ISBN 3 11 011626 X (Walter de Gruyter, 1994).

Dirichlet forms and Markov processes (Kodansha and North-Holland, 1980), by the first named author of the book under review, was a milestone in the theory of Markov processes. It became a major source for anyone wishing to understand the properties of Markov diffusion processes, and the main source for reversible processes, that is, those where the transition semigroup comprises self-adjoint operators.

Over the intervening 14 years, and in part stimulated by the original book, we have seen a number of developments. These have substantially underpinned the importance of the material in the book and encouraged many more to pick it up and read it. But these same developments, and a shifting emphasis towards infinite dimensional settings, have resulted in the bridge between the original text and current knowledge expanding, and a systematic redevelopment of the material has been required for some time.

The gap grew wide enough that two books which explained recent developments, N. Bouleau and F. Hirsch, *Dirichlet forms and analysis on Wiener space* (Walter de Gruyter, Berlin, 1991) and Röckner and Ma, *Introduction to the theory of (non-symmetric) Dirichlet forms* (Springer, Berlin, 1992), have appeared to general acclaim by appreciative readers. This reviewer also wrote a short survey in 1992.

So with all these new materials, why are we seeing a new version of the old favourite? Has it been worth waiting for? The answer has to be an emphatic yes. This is a beautifully written book, carefully put together, and as easy to read as one could expect. Although based loosely on the old text, much thought has been given to rewriting and reorganising the material exploiting the benefit of hindsight.

I would say at least half of the book has been rewritten with many new features. It will be essential reading for another 15 years. This is a carefully written and technically detailed book, bringing out the key points in a consistent manner. It supersedes the old book without expanding so dramatically as to become unwieldy.

I hope this book will stand the test of time and become an important influence on later generations. As a text for those wishing to give themselves a firm understanding of reversible Markov process theory which is rigorous and precise, but rich enough that it provides much of the machinery to work with the main models that might occur in finite and infinite dimensions, this is the book.

One word of warning. This is a relatively abstract book, with no compromises; this approach is justified by the many concrete examples fitting into the framework discussed here, and by the strength of some of the results. Reversibility exerts a very powerful influence on diffusion processes. To balance this abstraction, the reviewer feels that a young reader should come to this book with his own range of concrete processes (or putative self-adjoint elliptic operators), against which he can test and try the results he will find here.

TERRY LYONS

FROM BROWNIAN MOTION TO SCHRÖDINGER'S EQUATION
(Grundlehren der Mathematischen Wissenschaften 312)

By KAI LAI CHUNG and ZHONGXIN ZHAO: 287 pp., DM.148.—,
ISBN 3 540 57030 6 (Springer, 1995).

The title of this new book suggests a continuation of Chung's *Lectures from Markov processes to Brownian motion* which also appeared in this Springer series. That, however, applies only partly, as Chung himself indicates in the preface. Whereas Chung's previous textbook covers a large classical area of mathematics, the new book by Chung and Zhao is devoted essentially to one single, rather specific topic.

As the title indicates, the book deals with the connection between *Brownian motion* (P^x, X_t) and the *Schrödinger equation* $(\frac{1}{2}\Delta + q)u = 0$ on $\mathbb{R}^d$. More precisely, it deals with properties of *stopped* Brownian motion on a region $D \subset \mathbb{R}^d$ and its relation to the Schrödinger equation $(\frac{1}{2}\Delta + q)u = 0$ on D with Dirichlet boundary conditions. The case $D = \mathbb{R}^d$, usually regarded as most important, is in the background here—due to the credo of the authors: *Time must have a stop!*

Let us recall the well-known *Feynman–Kac* formula, which in the case of bounded q states that

$$u(\cdot) \mapsto P_t^q u(\cdot) = E^{\cdot} \left(\exp \left[\int_0^t q(X_s) ds \right] \cdot u(X_t) \right) \quad (1)$$

defines a semigroup whose generator is precisely the Schrödinger operator $\frac{1}{2}\Delta + q$. In general, this semigroup is no longer contracting, but it is bounded on each $L^p(\mathbb{R}^d)$, $1 \leq p \leq \infty$, by

$$\|P_t^q\|_{p,p} \leq \sup_x E^x \left(\exp \left[\int_0^t q(X_s) ds \right] \right) \leq \exp[\|q\|_{\infty} \cdot t].$$

An entirely different situation appears if in (1) the fixed time t is replaced by a

random time τ , for example, the first time τ_D at which Brownian motion exits a given bounded domain $D \subset \mathbb{R}^d$. Here a surprising phenomenon occurs. The expression

$$E^x \left(\exp \left[\int_0^{\tau_D} q(X_s) ds \right] \right), \quad (2)$$

which (regarded as function of the starting point x) is called the *gauge function*, is by no means always finite. This has to do neither with ‘singularities’ of q nor with ‘irregularities’ of D . For instance, for constant $q \equiv \alpha$ the gauge function is given by the expectation $E^x(\exp[\alpha \cdot \tau_D])$, which may be infinite in spite of the fact that, of course, $E^x(\tau_D) < \infty$. Indeed, there always exists a number $0 < \alpha_D < \infty$ such that

$$E^x(\exp[\alpha \cdot \tau_D]) < \infty \quad \text{and} \quad E^x(\exp[\alpha' \cdot \tau_D]) = \infty \quad (3)$$

for all $x \in D$ and $\alpha < \alpha_D \leq \alpha'$.

The *Gauge Theorem* of Chung and Rao from 1981 now states that the gauge function (2) is *either* bounded in D or identically $+\infty$. The relation to the Schrödinger equation is that the gauge function is bounded if and only if the spectrum of the Schrödinger operator $\frac{1}{2}\Delta + q$ on D with Dirichlet boundary condition is contained in the interval $(-\infty, 0)$.

This Gauge Theorem, together with its modifications and corollaries, is the main topic of the book under consideration. During the last 15 years, K. L. Chung has devoted most of his scientific interest to this field of mathematics. In the meantime, there exist many improvements and generalizations. The authors of the monograph in question restrict themselves to presenting the results in the case of Brownian motion on subsets of $\mathbb{R}^d$.

The first two chapters of the book serve as an *introduction to Brownian motion* and summarize results from *probabilistic potential theory* on $\mathbb{R}^d$. In Chapter 3, the *Kato class* is introduced, and results on Kato functions are derived which are essential for the sequel. The first culmination is reached in Chapter 4, where the *Gauge Theorem* mentioned above (together with various applications) is proven. The proof presented here is based on a *Harnack inequality* for local solutions of the Schrödinger equation $(\frac{1}{2}\Delta + q)u = 0$ in D .

Chapters 5–7 deal with generalizations of the Gauge Theorem, in particular with the so-called *Conditional Gauge Theorem*. The latter states that the conditional gauge function $E_y^x(\exp[\int_0^t q(X_s) ds])$, regarded as a function of the starting point $x \in D$ and of the terminal point $y \in \bar{D}$ of the conditional Brownian motion (P_y^x, X_t) in D , is either identically $+\infty$ or bounded. Actually, it is bounded if and only if the gauge function is bounded. The boundedness of the conditional gauge function entails many other properties. For instance, it implies that the Green functions (and as well the Poisson or Martin kernels) for the operators $\frac{1}{2}\Delta + q$ and $\frac{1}{2}\Delta$ are comparable on D .

The proof of the Conditional Gauge Theorem requires additional regularity assumptions on the boundary of D (for example, Lipschitz boundary). In this context, several nice and interesting results on conditional Brownian motion on Lipschitz domains are derived. Moreover, important estimates are derived for the classical Green function in such domains, for instance, the so-called 3G-Theorem.

In Chapter 8, the authors present several other developments and related problems. Among them is the question of whether the boundedness of the gauge function is stable under (‘small’) perturbations of the domain D or of the potential q .

Chapter 9, finally, deals with the one-dimensional case, which is excluded in the previous chapters.

Each chapter ends with historical notes and comments. Unfortunately, some important contributions (for example, those of Albeverio, Blanchard, Ma, Baxter, Dal Maso, Mosco, Feyel, de La Pradelle, Glover, Rao, Song) are omitted here as well as in the references.

The book is written very carefully and thoroughly. It contains a lot of interesting mathematics, and it is pleasantly readable. It is a nice book and a valuable contribution to mathematics.

KARL-THEODOR STURM

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